DESIGN PORTFOLIO

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Link to video demo

1 UX OF E-NOSES

A SENSORY EXTENSION

ROLE

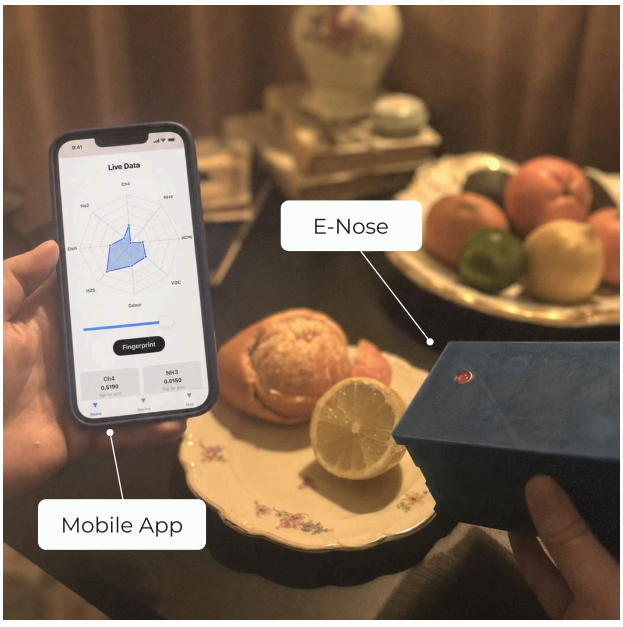
Project Lead

DURATION

6 months

TYPE

Individual



Link to video demo

ABOUT:

A Sensory Extension inquires into the future UX of Electronic Noses[1] for non-experts. The design process consists of two connected perspectives: an experimental E-Nose mockup and a React Native BLE application. The experiential prototyping of interactivity deepened the possibility of the technology without being limited by usability concerns. Subsequently, the app builds on the speculative results through a more practical approach.

TOOLS:

Electronics	React Native	BLE
MEMS	Soldering	Figma

[1] Electronic Nose is a sensor-based device that detects chemical compounds in the air. In this project, it is explored through smell-walking: a practice of moving through an environment while deliberately noticing smells. The E-Nose captures scent data as people walk, connecting measurable data with embodied experiences of noticing, interpreting, and reflecting on smell

THE EXPERIENCE

As the participant carries the E-Nose it responds to their movement. When they slow down, a calm haptic cue reminds them that the **device is present** and **ready for input.** Simultaneously, a continuous LED animation shifts with the internal data, allowing the participant to observe data changes without **relying on numerical values.**

PROTOTYPING

Walking pace grounded the interaction, as slowing down was the clearest observable sign that someone was beginning to smell. Using the Arduino Nano 33 BLE Sense Rev2's onboard BMI270/BMM150 sensors, I mapped pace data to LED and haptic feedback, allowing the mockup to give the **illusion of responding** to their smelling actions.

WHY IT MATTERS

The goal was to understand how an E-Nose could feel less like an unmanageable scientific tool that and more **like a sensory companion.** I focused on three qualities: dialogue, temporality, and transparency.



Early Prototypes

User Testing

CHANGES THROUGH TESTING

The mockup underwent qualitative testing with think-alouds. Participants experienced the interactive qualities as "another being," though for some, the haptic patterns felt hectic, **affecting their emotional relationship** to the device. The initial data feedback also did not fully reflect internal data shifts, so the final prototype introduced calmer haptics and more expressive visual feedback.



Experimental user flow

LEARNING OUTCOMES

Participants experienced the E-Nose as companion. Its expressive behaviour implied a sense of **mediated agency**. One participant described: "I had control, and it had control over me." This showed that the E-Nose was not only a data collection instrument, but an **interface for dialogue.**

The device influenced participants' perception of their own bodies, with one participant describing the experience as being **"one with the machine."**

This relationship **depended on attunement:** some participants felt directly connected to the vibrations, while others felt uncertain why they were there.

The design showed that when we design alongside people's existing activities, tools can affect our embodied actions and relationship to the world. This shifts the perspective from tools as **passive objects to active participants.**

PRACTICAL APPLICATION: MOBILE INTERFACE

The Sensory Extension work continued to more the developemnt of a React Native BLE application, connecint to a mobile via BLE. The interface was built for an in-lab E-Nose, that lacked live data communition. One of the core challenges was understanding how **intangible E-Nose data** could be communicated, and how this shapes **contextual user experiences.**

DESIGN QUESTIONS

The app is a **developing probe** for further field work and user testing. It opens questions around how the experience should be designed: should inputting sensor data happen through the E-Nose, phone interface, or through both? How much data is accesed depedning on context, and what type of data is meaningful?

PILOT TESTING INSIGHT

In pilot user testing, participants implied that both device and mobile application should provide forms of feedback, as taking out a phone can be **tedious and distracting.**

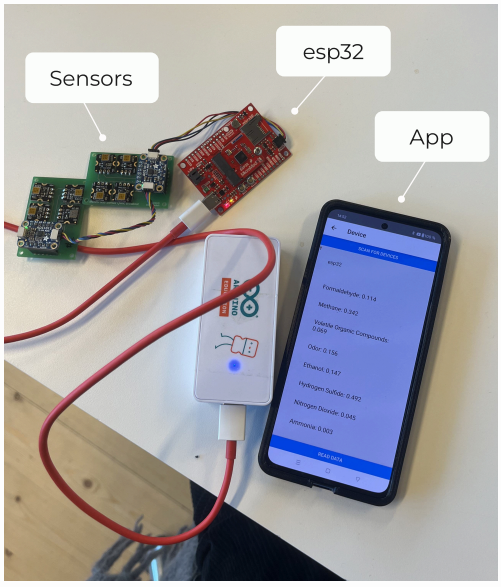
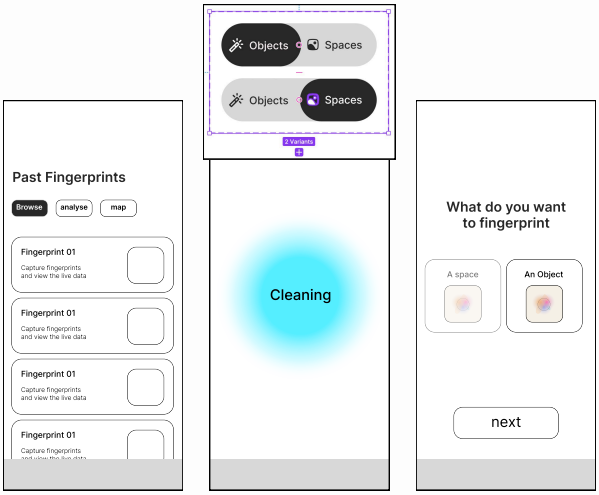


Diagram: 8 MEMS sensors → esp32 → BLE → Mobile Phone



Ideation & Mockups

TECHNICAL BUILD

The application was built **cross-platform** with React Native. The hardware was programmed with **BLE infrastructure.**

THE EXPERIENCE

The user has direct access to sensor data through a radial chart and numerical values. The radial chart provides an overview of the data samples in relation to each other, allowing **users to visually identify patterns** without depending on numerical data. This supports immediate comparison and makes the interface more **accessible for expert and non-expert users.**

FUTURE DIRECTION

The project is ongoing, and future research within academia, citizen science, and environmental monitoring will provide further insight into the different user and environmental circumstances that may change the interface. The ultimate goal is an **adaptive interface** that adjusts according to context and user.



Link to video demo

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OIKOTONICA

AN ADAPTIVE MIDI CONTROLLER

ROLE

Project Lead

DURATION

5 weeks

TYPE

Team



ABOUT:

Oikotonica, a sensor-based MIDI controller that changes with its environment. It invites musicians to connect with surrounding ecologies through adaptive sonification. By converting live environmental data into MIDI signals, Oikotonica creates ever-evolving, unrepeatable interactions.

TOOLS:

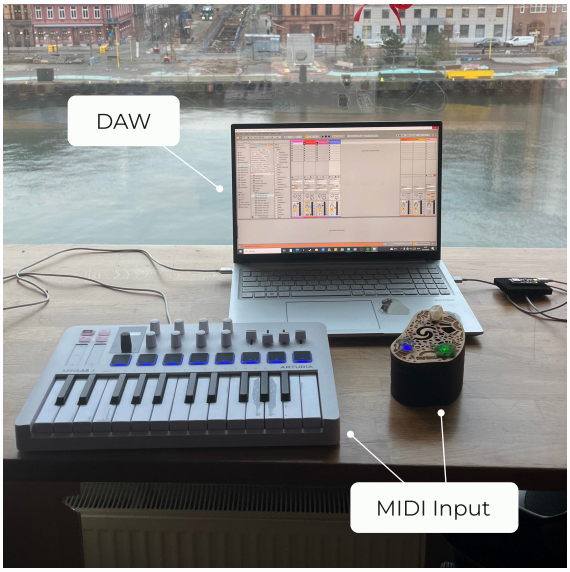
Electronics	Arduino (C++)	BLE
MIDI CC	Sensors	Soldering

THE EXPERIENCE

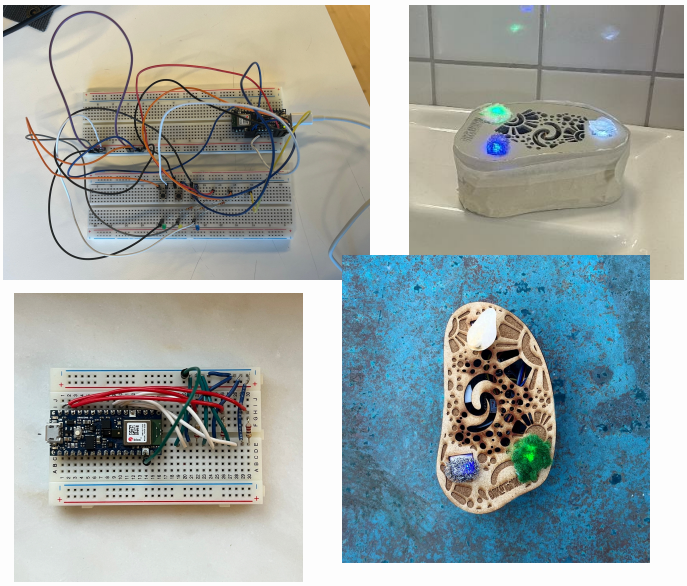
The instrument is handheld and meant to merge with **existing musical setups.** Musicians can map the sensors to their preferences in their DAW, either to specific controls or sounds. The instrument uses the surrounding **environment as input**, while also allowing bodily and gestural manipulation for more control. The buttons are texturized, allowing participants to identify controls by touch only. Oikotonica **shapeshifts across space and time,** becoming an **agnostic interface** that can support live performances or help artists iterate while producing.

PROTOTYPING

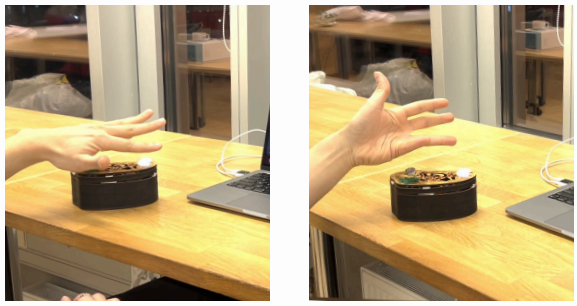
Using two Arduino Nano 33 BLE Sense boards, three onboard sensors turned into BLE-connected MIDI control inputs: humidity through the HTS221 sensor, light through the APDS-9960, and ambient sound through the PDM microphone. Each sensor wast ransformed into MIDI parameters. Each was activated through its own button and naturally evolved into different **temporal qualities,** humidity as slow, light as medium, and sound as fast.



Use case example



Prototype timeline



Gestural interactions with light sensor

WHY IT MATTERS

Musical instruments often enable replicable musical experiences, remaining uniform regardless of location, audience, or context. This project embraces instability by incorporating the environment's unpredictable qualities into live sound alteration. Through adaptive sonification and the interaction of playing a musical instrument, the project explores **the intersection of technology, environmental awareness, and artistic expression.**

CHANGES THROUGH TESTING

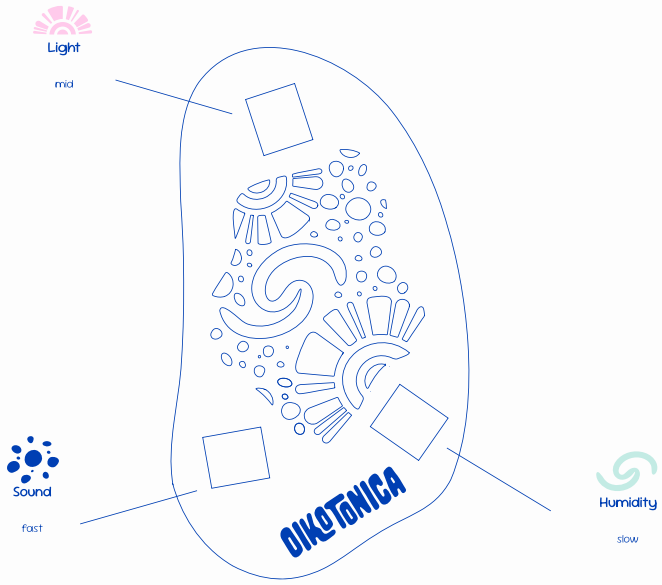
The instrument underwent qualitative testing with musicians at different levels. They wanted to hear the unmodified MIDI parameters as a source of comparison, a hold button function was implemented that turned off the sensors while pressed. Participants indicated that transitions were too abrupt and suggested adding an easing effect, as well as dedicated **software for more control** in live performance and practice contexts.

LEARNING OUTCOMES

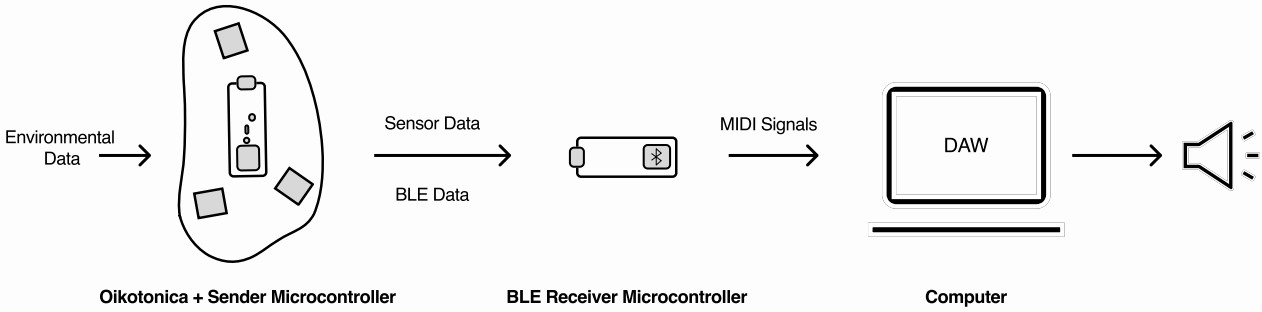
Meaningful connections with the instrument required **an initial relationship** with how the instrument responded to surrounding environments.

Even with rough prototypes and exposed microcontrollers, participants were physically manipulating the sensors. This behaviour continued into the high-fidelity prototype, suggesting that **bodily input was a significant source** when playing, rather than allocating control entirely to an outside source: the environment.

The project suggests that the unpredictable qualities of the environment, the body, and sensors can blend into **new forms of nuanced play**, exploring the future of agnostic and shapeshifting musical interfaces.



Sensor mapping



Diagram

3

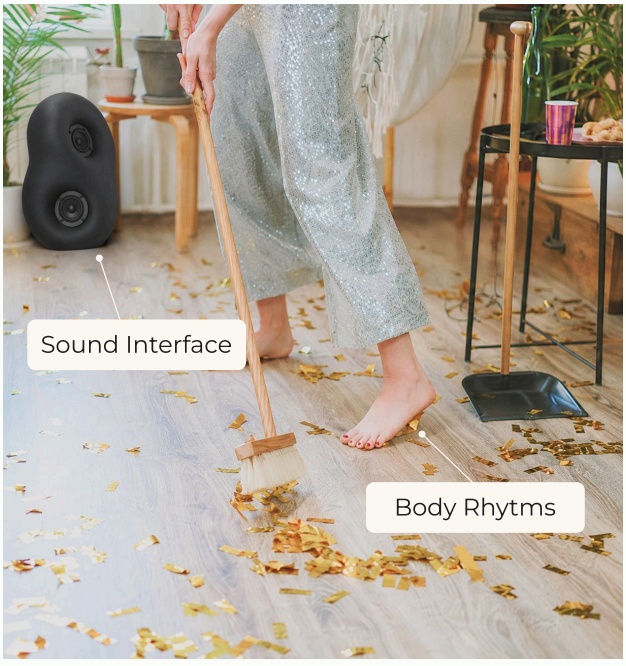
BODILY RHYTMS

ADAPTIVE SOUND INTERFACE

ROLE
Project Lead

DURATION
(ongoing)

TYPE
Individual



ABOUT:

This project explores how an adaptive ambient sound interface can tune into domestic and bodily rhythms, making daily moments feel more magical. Taking inspiration from generative music, sound is treated less as a notification and more as an aesthetic and interactive presence that evolves with the body and the task. The project aims to reveal emergent interactive patterns that can be applied to the novel design space of Augmented Audio Reality.

TOOLS:

Electronics Supercollidor Arduino (C++)

Sensors

DESIGN SPACE

The design space inquires in the potential sound and music that anticipates and evolves alongside our **activities and bodily rhythms.** Rhythms are an inherent part of our daily life: circadian rhythms, pules, walking, biking, cleaning, etc.

Domestic settings are full of these existing body rhythms, but they often go unnoticed as mundane routines. These actions are not performed like exercise or dance, but they still have tempo, rhythm, and variation, **serving as material for generative sound.**

The project asks what happens when these existing rhythms are sensed, translated, and reflected back as **agentic ambient music**. It explores how this may influence our actions and perception of our tasks, specifically focusing on the intersection between foreground and background presence.

INITIAL CONCEPT

The original idea started from moods, and how a home could adjust to or reflect a certain moodiness through sound. From there, the project moved toward generative music and bodily rhythms, inspired by **SuperCollider and Brian Eno's generative music.**

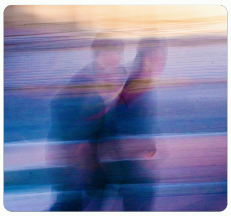
PROTOTYPING

The design uses the XIAO 60GHz mmWave Human Breathing and Heartbeat Sensor (MR60BHA2) to capture bodily rhythms. This data is transformed into SuperCollider parameters.

CURRENT STATE

Initial surveys were conducted to probe the territory, and further work will involve having the prototype exist with people for a longer period of time. The design space is still emerging through exploration, but the project tests the **potential of augmented audio reality** within everyday activities, domestic settings, and already existing bodily rhythms.

The results of this design can inform future work within augmented reality and **invisible adaptive interfaces** that are closely connected to our bodies.



Body Rhythms



Sensing Interface



Adaptive Sonic Output



Generative Sound System

Diagram

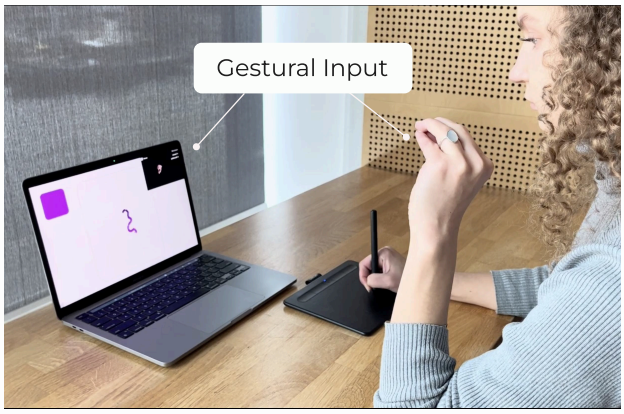
4 MICRO-ACTIONS

THE OTHER HAND

ROLE
Project Lead

DURATION
3 weeks

TYPE
Pair



Link to video demo

ABOUT:

Through object detection technology, our physical bodies can enrich digital experiences by serving as additional inputs. Hands as +1 combines traditional cursors with gestures in digital drawing. The dominant hand draws with a cursor/tablet, while the other acts as a simultaneous color adjuster—allowing multiple actions without tool switching. This mirrors real-world tasks, where both hands actively contribute to a workflow.

TOOLS:

Mediapipe JavaScript HTML/CSS

THE EXPERIENCE

With your non-dominant hand, you entrap the hue by tightening your grip. Tilting or changing the angle of that hand then modulates the shade. With your dominant hand, you create temporary colour swatches, which disappear once you release the current colour.

The design emphasizes the in-between state of drawing and choosing a colour. It creates **a small space for discovery:** a time window to search, try, and adjust colour without leaving the flow.

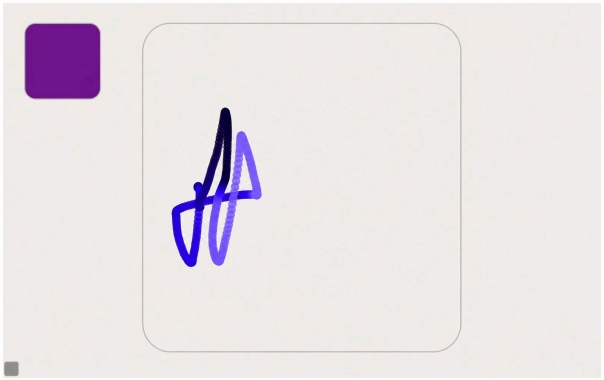
The **same action and space** take on different affordances, removing the need to toggle between sliders, tools, or panels. Instead, you see the changes directly through swatching, making colour selection part of the act of drawing rather than a separate activity.



Close up of hand gestures

WHY IT MATTERS

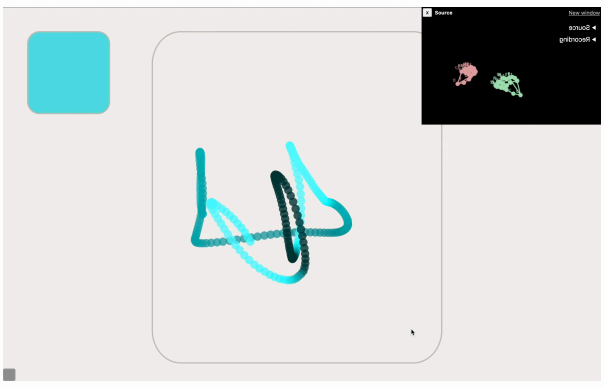
The body can serve as a rich input in digital interactions, where so much of it goes unused. By exploring gestural interaction within the **micro-context** of existing activities, we can open up a conceptual space and further develop the tools we already use toward more **expressive and sophisticated interactions.** Through this, we also explore what it means to create an activity **within an activity through micro-actions.**



GUI mockup of a drawing interface

PROTOTYPING

Gestures were mapped to pointer events using a webcam and MediaPipe. A user interface imitating a drawing environment was created to explore the concept.



Pilot phase of both hands as gestural input

THANK YOU.